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October 3, 2026
ECE 535
Project Proposal

Robotic Manipulator with Visual Language Action Models (VLA)

Motivation:

- Robotics is the next frontier of AI and the individual nature of robotic agents means that local inference will be highly important to complete daily tasks in any environment. Getting VLA models small and effective enough to run locally furthers the capabilities of robots to complete otherwise difficult to automate tasks without the need for cloud inference.

Design Goals:

- Language-driven control: The robot arm's trajectory is controlled by natural-language
- The compressed 7B VLA runs in less than 16 GB of memory
- The compressed model completes pick-and-place tasks with a success rate close to the full-size model
- The model makes decisions fast enough to control the arm smoothly
- The model handles different objects, colors, and target locations.

Deliverables:

- Compressed 7B VLA model to 16GB (in simulation)
- Simulated pick and place environment in a robotic simulator with a VLA compressed model
 - Isaac Sim tabletop scene with a robot arm and objects to manipulate
 - The arm follows language instructions using the compressed model
 - Simulation code, evaluation results, and a demo video

System Blocks:

1. The user gives natural-language commands.
2. A robot arm, a table, and objects for pick-and-place.
3. The model server receives the camera image and instruction and runs the VLA, either on the same machine as the simulator or on a separate one.
4. The compressed VLA model turns the image and instruction into the next arm movement.
5. The arm controller converts that movement into joint commands in the simulator.
6. The evaluator measures task success, memory use, and decision speed.

Hardware/Software Requirements:

- Nvidia IsaacSim (<https://github.com/isaac-sim/IsaacSim>)
- Python
- <16 GB memory size

- GitHub for version control

Team Members Responsibilities:

- Wyatt
 - Install Isaac Sim, NVIDIA drivers, CUDA, and the Python environment on the GPU machine
 - Build the Isaac Sim pick-and-place scene, camera, arm controller, and simulation loop. Run all simulation trials and record the demo video
 - Run the final compressed models on the GPU machine and measure their memory use and speed in the full system.
- Isabel
 - Write the compression scripts (e.g., 8-bit and 4-bit quantization). Choose how instructions are worded for the model, and analyze the memory, speed, and success-rate results.
 - Build the model server and the link between the simulator and the model.

Project Timeline:

- Week 1 (Oct 3–9): Create the GitHub repo, assign roles, read the references, arrange GPU access.
- Week 2 (Oct 10–16): Install Isaac Sim and load the robot arm scene, run the full-size VLA on sample images.
- Week 3 (Oct 17–23): Build the pick-and-place scene and camera, begin compressing the model.
- Week 4 (Oct 24–30): Model server working (instruction → arm moves).
- Week 5 (Oct 31–Nov 6): Finish the VLA explanation: measure memory and speed for the compressed models.
- Week 6 (Nov 7–13): Build the evaluator, run trials across different instructions and object positions.
- Week 7 (Nov 14–20): Improve the compression approach and simulation setup.
- Week 8 (Nov 21–27): Final comparison of the full-size and compressed models.
- Week 9 (Nov 28–Dec 4): Finalize the demo and the code.
- Week 10 (Dec 5–11): Final report and presentation.

References:

- An Open-Source Vision-Language-Action Model. (<https://arxiv.org/pdf/2406.09246>)
- Rethinking the Practicality of Vision-language-action Model: A Comprehensive Benchmark and An Improved Baseline (<https://arxiv.org/pdf/2602.22663>) (<https://github.com/OpenHelix-Team/LLaVA-VLA>)
- <https://docs.nvidia.com/learning/physical-ai/getting-started-with-isaac-for-healthcare/latest/training-healthcare-robots-from-scratch/04-model-flywheel/03-deploy.html>

Team Roles:

Isabel: Algorithm Design & Networking Lead

Wyatt: Setup & Software Lead